Bei Xiao

Guangzhou, China | beixiao0507@gmail.com
GitHub: Claire-bei / OCTA-Retinal-Vessel-Demo

RESEARCH PROFILE

Ophthalmology researcher with a professional master's degree in clinical medicine (ophthalmology) and first-author publications in quantitative retinal and choroidal imaging. Experienced in ultra-widefield SS-OCTA, vascular phenotyping, image-quality assessment, and clinical data analysis. Developed Python/OpenCV image-processing workflows and implemented an initial PyTorch/ResNet18 fundus-age baseline with patient-level dataset splits and ongoing validation image-quality review.

RESEARCH INTERESTS

AI-assisted analysis of multimodal retinal imaging; quantitative imaging biomarkers; retinal and macular disease phenotyping; image-based prediction of disease progression.

EDUCATION

Professional Master's Degree in Clinical Medicine (Ophthalmology)

Southern Medical University, Guangzhou, China | Sep 2021-Jun 2024

Bachelor of Medicine in Clinical Medicine

Hubei University of Science and Technology, Xianning, China | Sep 2016-Jun 2021

RESEARCH EXPERIENCE

Computational Reanalysis of Human CNV Transcriptomic Data

May 2026-present

Independently reanalysed public choroidal neovascularization datasets using R: preprocessing, PCA, differential expression, and functional enrichment. Interpreted candidate immune and extracellular-matrix remodelling signals. Prepared and submitted the manuscript as first and corresponding author.

Quantitative Retinal and Choroidal Imaging in Pachychoroid Spectrum Diseases

Master's research | Sep 2021-Jun 2024

Conducted observational research using ultra-widefield SS-OCTA. Performed image-quality checks, segmentation review, retinal/choroidal measurements, and vascular phenotyping. Characterised pachyvessels, vortex vein dilatation, and choroidal venous drainage patterns. Analysed clinical imaging associations and contributed to study design, statistical interpretation, and first-author publications.

COMPUTATIONAL PROJECTS

FundusAgeAI: Fundus-Photograph Age Prediction Demo

2026-present | Ongoing

Implemented an initial PyTorch/ResNet18 baseline with patient-level training, validation, and test splits. Conducting age-stratified, blinded manual image-quality review and exploring image-quality effects on prediction performance.

OCTA Image-Processing Demo

Python / OpenCV | Exploratory project

Developed scripts for OCTA enhancement, binarisation, vessel skeletonisation, and vessel-width estimation. View repository.

ACADEMIC APPOINTMENTS

Guangzhou Xinhua University | Faculty Member | Sep 2026-present
Department of Optometry, School of Medicine and Health. Teaching and coordinating optometry courses and practical sessions.

Guangzhou Huali College | Teaching Faculty | Nov 2025-Jul 2026
Taught medical-related courses and participated in undergraduate teaching activities.

Bei Xiao

Publications & research skills

PUBLISHED ARTICLES

1. **Xiao B**, Song YP, Yan M, Ye Y, Huang Z. Quantitative Analysis of Choroidal Vascular Structures and Anatomical Changes in Pachychoroid Spectrum Diseases Using Ultra-Widefield SS-OCTA. *Scientific Reports. 2025;15:344.* 10.1038/s41598-024-82745-5

2. **Xiao B**, Yan M, Song YP, Ye Y, Huang Z. Quantitative Assessment of Choroidal Parameters and Retinal Thickness in Central Serous Chorioretinopathy Using Ultra-Widefield Swept-Source Optical Coherence Tomography: A Cross-Sectional Study. *BMC Ophthalmology. 2024;24:176.* 10.1186/s12886-024-03405-w

3. **Xiao B**, Song YP, Ye Y, Huang Z, Yan M. Imaging Characteristics of Affected and Fellow Eyes in Patients with Central Serous Chorioretinopathy at Different Ages. *International Eye Science. 2024;24(6):937-942.* 10.3980/j.issn.1672-5123.2024.6.19

4. **Xiao B**, Song YP, Yan M, Ye Y, Huang Z. A Comparative Study of Choroidal Parameters in the Fellow Eyes of Patients with Pachychoroid Neovasculopathy and Central Serous Chorioretinopathy. *Recent Advances in Ophthalmology. 2024;44(5):376-381.* 10.13389/j.cnki.rao.2024.0073

5. **Xiao B**, Song YP, Yan M, Ye Y, Huang Z. The role of vortex veins in the pathogenesis of central serous chorioretinopathy. *Chinese Journal of Ocular Fundus Diseases. 2024;40(4):319-323.* 10.3760/cma.j.cn511434-20231024-00430

6. **Xiao B**, Song YP, Ye Y, Huang Z, Yan M. Association Between Vortex Vein Dilatation Patterns and Choroidal Thickness Changes in Patients with Central Serous Chorioretinopathy. *Chinese Journal of Ocular Fundus Diseases. 2024;40(5):360-365.* SinoMed article record: 2024322835

7. **Xiao B**, Song YP, Yan M, Ye Y, Huang Z, Chen C, Deng YM. Observation on the Efficacy of Subthreshold Micropulse Laser Photocoagulation in the Treatment of Central Serous Chorioretinopathy Based on Multimodal Imaging Classification. *Recent Advances in Ophthalmology. 2024;44(7):531-535.* 10.13389/j.cnki.rao.2024.0102

MANUSCRIPTS UNDER SUBMISSION

Xiao B, Lu J. AI-Based Retinal Oculomics: A Translational Evidence-Chain Review from Ocular Detection to Systemic Risk Stratification. *Technology and Health Care.* Revised manuscript submitted. First and corresponding author.

Xiao B, Lu J, Zhang X. Computational Reanalysis of Human Choroidal Neovascularization Transcriptomes Identifies Immune-Remodeling Candidate Signals. *Computational Biology and Chemistry.* Submitted. First and corresponding author.

TECHNICAL SKILLS

Imaging: Ultra-widefield SS-OCTA/OCT; image QC; segmentation review; retinal/choroidal measurements.
Programming: Python, OpenCV, PyTorch/ResNet18, NumPy, pandas, Matplotlib, Jupyter.
Analysis: R, SPSS, Stata, GraphPad Prism; regression, PCA, differential expression, functional enrichment.
Laboratory collaborations: Hands-on qPCR, Western blotting, RNA extraction, cell culture, and transfection during master's laboratory collaborations.
Languages: Mandarin Chinese (native); English (IELTS Academic overall 6.5).